

AMD (AMD US)

Advancing AI Takeaways – Mgmt Raising TAM Aggressively

Maintained Buy
Target price \$640

AMD raises TAM for AI accelerator: The company now forecasts AI accelerator market to reach \$1.4tn by 2030, up from its prior datacenter silicon TAM estimate of \$1tn last year, driven by agentic AI workloads pushing the demand for reasoning. This is consistent with our constructive stance that the AI cycle despite the recent market unwind (see our *July Edition: Beyond the Unwind on July 14*). We raise our AI accelerator TAM forecast to \$1tn by 2028.

AMD raised TAM for server CPU: The company now forecasts server CPU market to reach over \$220bn by 2030, driven by agentic AI (~50% of TAM) where CPU:GPU ratio potentially tightening toward 1:1, datacenter upgrade to high-density CPUs to free up power and rack space, and enterprise AI adoption to deploy cost-effective computing.

MI455X a big leap, followed by MI500: MI455X delivers much better performance and cost profile, narrowing the gap with Nvidia, with 34x higher throughput and 18x lower token cost at highest concurrency than MI355X, where MI355X already outperformed Nvidia's B200 in some LLM inference workloads with 1.2-1.3x higher throughput and 50% lower cost. For MI500, it marks AMD's transition toward optical scale-up interconnects, which we expect to be ahead of Rubin Ultra in hardware due to a 4-die package, larger HBM capacity and larger first-layer scale up domain. Helios rack provides 15% higher compute performance, 50% more HBM4 memory capacity and scale-out bandwidth, and 10% higher throughput under high concurrency vs. Vera Rubin NVL72. Meanwhile, we believe ROCm is also a key watch, where users can migrate from CUDA while preserving 75% of existing CUDA code.

Venice CPU the most capable for AI workloads: Venice family offers a workload-optimized CPU portfolio spanning agentic AI, cloud, enterprise, and HPC applications, delivering 2.2x/2.0x higher throughput and 1.2x/1.3x better per core performance vs. Vera/Xeon 6. Underpinned by Venice's superior spec and performance, we expect ~300k CoWoS builds in 2027, implying 5-5.5m units, which should lift datacenter CPU ASPs by 30–35% YoY in 2027, assuming no like-for-like price increases.

Maintain Buy with TP at \$640: Following AMD's announcement on partnership with Anthropic on July 22 and the Advancing AI event on July 23 where AMD raised TAM forecast aggressively, we reiterate our bullish view on AMD and reiterate Buy with TP of \$640.

Risks: 1) AI demand slowdown; 2) new product delay; 3) peer competition.

Profit forecast

	2024	2025	2026E	2027E	2028E
Revenue (m)	25,785	34,639	52,324	82,247	113,313
Revenue YoY (%)	13.7	34.3	51.1	57.2	37.8
Net profit (m)	5,420	6,831	13,145	23,988	35,213
Net profit YoY (%)	26.0	26.0	92.4	82.5	46.8
EPS (\$)	3.3	4.2	7.9	14.2	20.2
P/E	107.4	85.1	44.8	25.0	17.6
ROE (%)	9.4	10.8	17.7	25.5	29.4

Source: Company data, GF Securities (Hong Kong) Brokerage.

Jeff Pu, CFA

SFC CE No. BNO719
jeffpu@gfgroup.com.hk

Henry Huang

SFC CE No. BYA749
henryhuang@gfgroup.com.hk

Advancing AI Keynote Takeaways

AMD raised TAM estimates for AI accelerator and CPU

During the Advancing AI event, the company forecasts AI accelerator market to reach \$1.4tn by 2030 and server CPU market to exceed \$220bn (with 50% of TAM from agentic AI) given CPU:GPU ratio potentially tightening to 1:1, up from its prior datacenter silicon TAM estimate of \$1tn last year, and overall TAM growing from ~\$365bn in 2025 to ~\$2tn in 2030, with ~40% CAGR. Mgmt attributed the upward revision to accelerating agentic AI, with monthly token generation reaching 35tn, ~160x higher than two years ago.

Overall, these are consistent with our constructive stance that the AI cycle remains intact despite the recent market unwind (see our *June Edition: Constructive Amid Volatility on June 16* and *July Edition: Beyond the Unwind on July 14*).

We raise our AI accelerator TAM forecast to \$1tn by 2028

Per our bottom-up AI accelerator model in which we integrate our forecast on Nvidia, AMD and major ASIC estimates, we now forecast the AI accelerator to grow from \$418bn in 2026 to \$1tn in 2028.

Figure 1: AI accelerator TAM forecast

	2025E	2026E	2027E	2028E
GPGPU (Units m)				
NVIDIA	6.0	9.4	11.0	10.1
AMD	0.5	0.8	1.1	1.3
Others	1.2	1.5	2.5	3.0
Total	7.7	11.7	14.6	14.5
ASIC (Units m)				
Google	2.9	4.8	9.3	12.3
AWS	1.6	2.5	3.1	2.3
Meta	0.2	0.3	0.4	0.4
MSFT	0.1	0.2	0.3	0.6
Other ASICs	0.8	1.0	1.5	3.0
Total	5.5	8.8	14.6	18.5
Market (USD bn)				
NVIDIA	163	330	521	703
Total GPGPU	178	358	575	780
Total ASIC	29	60	149	223
Total Market	208	418	724	1002

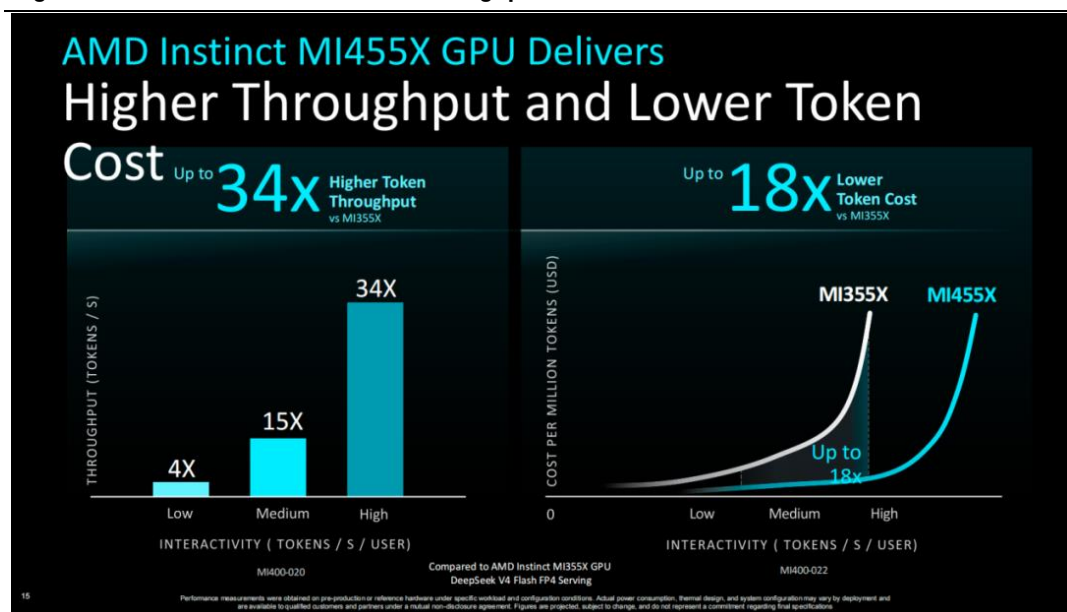
Sources: Company data, GF Securities (Hong Kong) Brokerage

MI455X to deliver much better performance and cost profile

AMD CEO Lisa Su showcased third-party benchmark results, demonstrating MI355X outperformed Nvidia's B200 in some LLM inference workloads, achieving 1.2-1.3x higher throughput in GPT-OSS-120B financial analysis and Qwen3-Next-80B interactive retrieval, while lowering cost per document by over 50%.

For MI455X, its performance further exceeds MI355X with 34x higher throughput and 18x lower token cost at highest concurrency, alongside 1.5x larger memory capacity, 2.9x higher bandwidth, and 4x higher peak MXFP8 compared with MI355X. We believe MI455X's lower token cost leaves AMD well positioned as the market shifting toward inference workloads, driving compelling customer ROI while narrowing the hardware gap with Nvidia, as said in our previous AMD report *Advancing AI to Underscore Enhanced CPU/GPU Outlook on July 21*.

Figure 2: AMD MI455X vs. MI355X at throughput and cost



Sources: Company data, GF Securities (Hong Kong) Brokerage

MI500 transition to optical scale-up

CEO highlighted three key features of MI500: new HBM, larger scale-up domain, and optical interconnects. Scheduled for commercial shipment in 2027, AMD targets 2,000x increase in throughput over a 4-year horizon to meet the extreme memory bandwidth and compute demands of multi-trillion parameter agentic AI workloads. MI500 marks AMD's pivotal transition toward optical scale-up interconnects, breaking traditional bandwidth distance and thermal constraints while leveraging open Ethernet/UALink targeting ~4000 Gbps data center interconnect speeds. We expect MI500 ahead of Rubin Ultra in hardware due to a 4-die package, larger HBM capacity and larger first-layer scale up domain.

Figure 3: AMD's AI GPU roadmap

	MI250	MI300A	MI300X	MI325X	MI355X	MI455X	MI550X
Launch	4Q21	4Q23	4Q23	4Q24	3Q25	3Q26	3Q27
GPU	CDNA2	CDNA3	CDNA3	CDNA3	CDNA4	CDNA5	CDNA6
Front-end node	N6	N5	N5x8, I/O N6x4	N5x8, I/O N6x2	N3Px8, I/O N6x2	N2x4, I/O N3x2	N2x8, I/O N3x4
Back-end process	CoWoS-S	SoIC+ CoWoS-S	SoIC+ CoWoS-S	SoIC+ CoWoS-S	SoIC+ CoWoS-S	SoIC+ CoWoS-L	SoIC+ CoWoS-L
Transistor (bn)	58.2	146	153	153	185	320	
Scale-up (uni-directional)	50GB/s	64GB/s	64GB/s	64GB/s	76.8GB/s	1700GB/s	CPO (optional)
HBM (GB)	HBM2e, 128GB	HBM3, 128GB	HBM3, 192GB	HBM3e, 256GB	HBM3e, 288GB	HBM4, 432GB	HBM4e, 768GB
Memory Bandwidth (TB/s)	3.2	5.3	5.3	6	8	19.6	
TDP	560W	750W	750W	1000W	1400W	2300W	

Sources: Company data, GF Securities (Hong Kong) Brokerage

Venice to be the most capable CPU for AI workloads

CEO indicated that datacenter is evolving toward a multi-dimension agentic AI architecture, each placing distinct requirements for CPUs. Agent sandbox CPUs orchestrate AI agents, execute AI-generated code, and manage non-GPU I/O; AI host CPUs keep GPUs fed with data; General CPUs continue to run databases and storage workloads. As such, Venice family offers a workload-optimized CPU portfolio spanning agentic AI, cloud, enterprise, and HPC applications, outlining several variants, including the 256-core Venice HF for agentic sandboxes and a 128-core version for general-purpose enterprise workloads.

Performance wise, AMD demonstrated that Venice delivers up to 1.8x the performance of Turin, doubling memory and I/O bandwidth, representing one of the largest generational gains in EPYC franchise. **Compared with Nvidia's Vera CPU and Intel's Xeon 6, AMD highlighted 2.2x/2.0x higher throughput vs. Vera/Xeon 6, and 1.2x/1.3x better per core performance than Vera/Xeon 6.** Underpinned by Venice's superior spec and performance, we expect ~300k CoWoS builds in 2027, implying 5-5.5m units, which should lift datacenter CPU ASPs by 30 – 35% YoY in 2027, assuming no like-for-like price increases.

Figure 4: AMD Venice vs. Nvidia Vera and Intel Xeon 6



Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 5: Major server CPU comparison

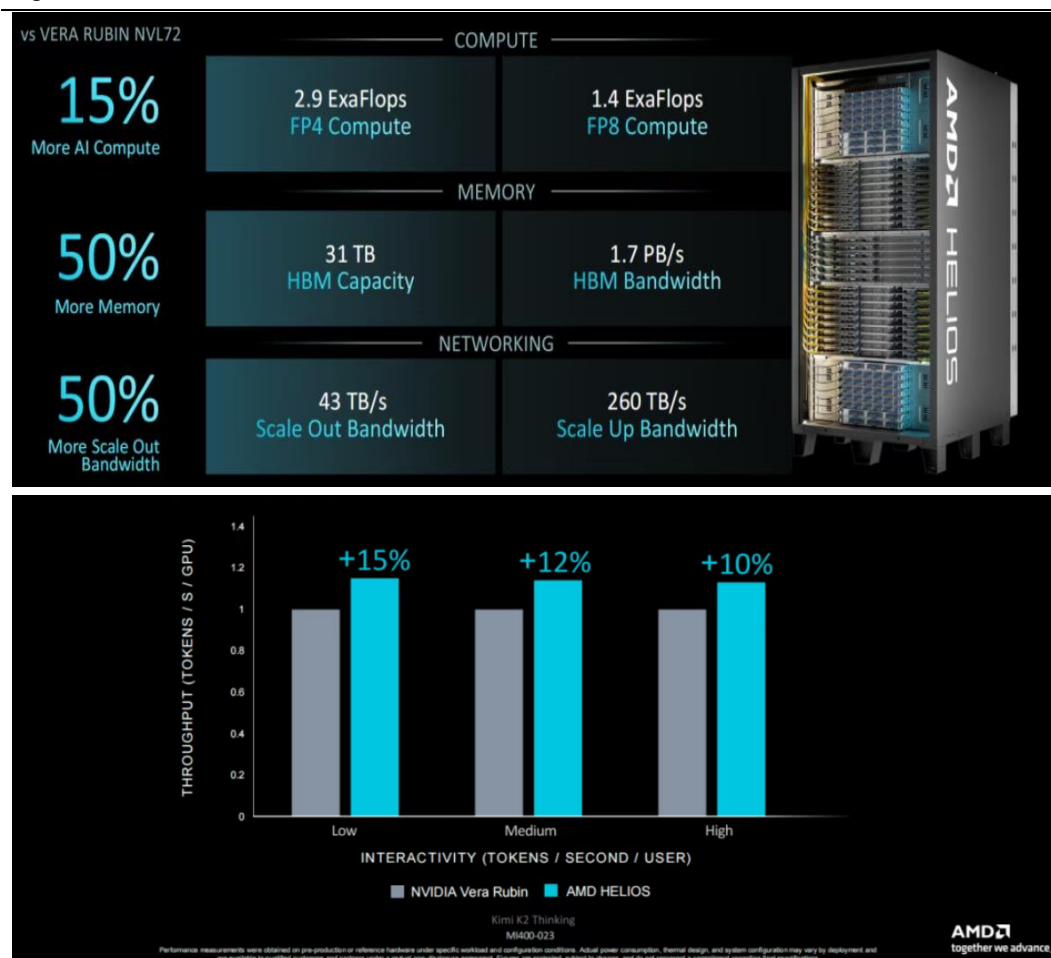
	Google Axion 2	AWS Graviton5	ARM AGI CPU	NVIDIA Vera	Intel Granite Rapids	AMD Turin	AMD Venice	Intel Clearwater Forest
Architecture	Neoverse N3	Neoverse V3	Neoverse V3	Olympus	Redwood Cove	Zen 5	Zen 6	Darkmont
Total SRAM	200 MB	400 MB	640 MB	338 MB	800 MB	1,248 MB	2,300 MB	1,152 MB
Max Core Count	72 vCPUs	192 Cores	136 Cores	88 Cores	128 Cores	96 Cores	256 Cores	288 Cores
Peak Clock	3.5 GHz	3.7 GHz	3.7 GHz	3.5 GHz	3.9 GHz	5.0 GHz	>5.0 GHz	~3.5 GHz
DRAM Capacity	768 GB (DDR5)	1.5 TB (DDR5)	6 TB (DDR5)	1.5 TB (LPDDR5X)	3 TB (DDR5)	6 TB (DDR5)	12 TB (DDR6)	3 TB (DDR5)
TDP	250W	300W	300W	200W	500W	450W	500W-700W+	500W
Process Node	TSMC N3	TSMC N3	TSMC N3	TSMC N3	Intel 3	TSMC N3/N4	TSMC N2	Intel 18A

Sources: Company data, GF Securities (Hong Kong) Brokerage

Helios rack the fully co-designed rack-scale solution

AMD positioned Helios as industry's highest-performance AI rack for training and serving frontier models at scale. Integrating MI455X GPUs, Venice CPUs, ROCm software stack, and Pensando networking, Helios is designed to deliver exaflop-scale AI performance. Compared to Nvidia's Vera Rubin NVL72, AMD claims Helios provides 15% higher compute performance, 50% more HBM4 memory capacity and scale-out bandwidth, and 10% higher throughput under high concurrency workloads. Besides hardware performance, we believe ROCm is also a key watch, where users can migrate from Nvidia's CUDA to AMD's ROCm while preserving on average 75% of existing CUDA code, and collaborations with Anthropic's Claude and OpenAI's Codex improve developer accessibility and optimization productivity and platform usability, making users easier to switch the system. Helios rack is expected to start commercial shipments from 3Q26 and ramp into 4Q26 and 1H27.

Figure 6: AMD Helios vs. Nvidia Vera Rubin NVL72



Sources: Company data, GF Securities (Hong Kong) Brokerage

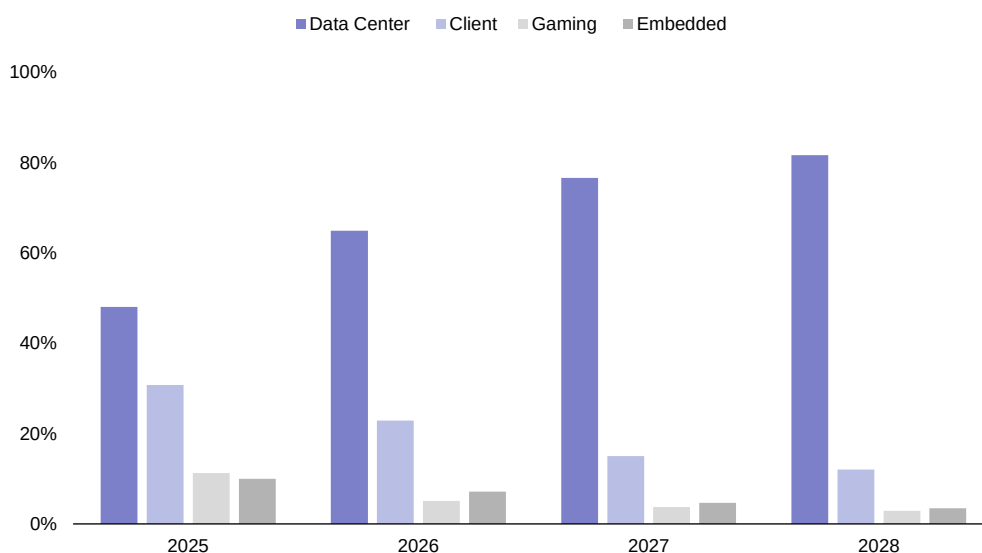
Datacenter gains traction while client business keeps pace

We forecast AMD's datacenter revenue to be \$34bn/63bn/31bn in 2026-2028E, representing 65%/77%/82% of total sales. However, we believe AMD's client business keeps pace as the company showed that it wants to serve more than hyperscalers, introducing products for enterprise servers, personal AI machines and physical AI at Advancing AI, where physical AI deployments could be a watch.

For enterprise AI, MI350P is designed for enterprise data centers that do not want to upgrade facilities, which can support models with 260bn parameters on a single GPU; for personal AI, Ryzen AI Halo and Gorgon Halo supporting 200bn/300bn parameters models will appear in laptops, workstations, mini PCs and developer systems, working with Hugging Face to optimize Halo for open models and agentic workflows.

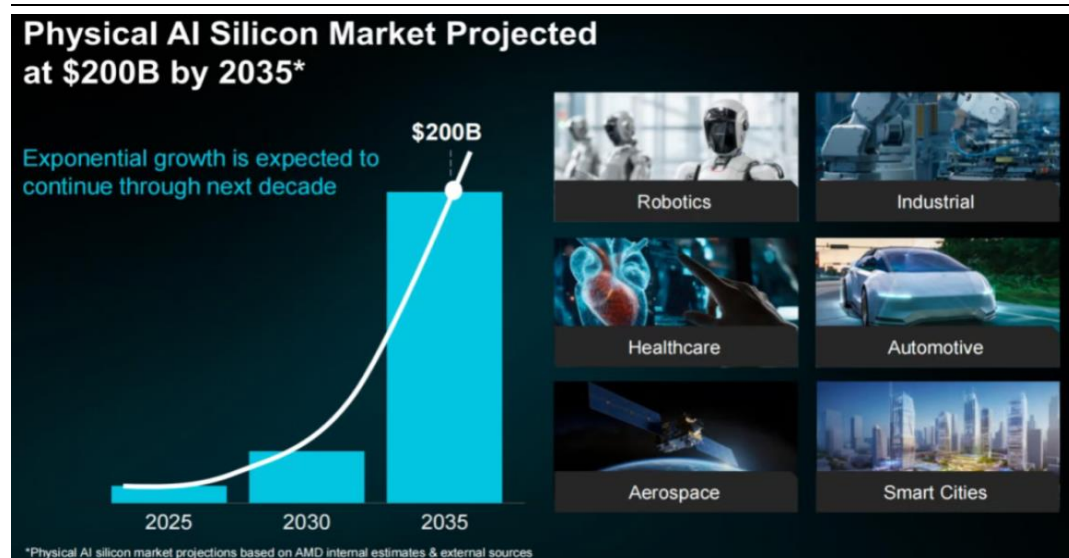
For physical AI, AMD indicated the potential of physical AI and forecasted its TAM to reach \$200bn in 2035. Ergo, AMD introduced Kria AI, a system-on-module powered by the Ryzen AI Embedded X100 that integrates CPU, GPU, NPU, and unified memory into a compact, open-standard platform. According to AMD, Kria AI delivers 3.4x better real-time performance than Nvidia's Jetson T5000, while supporting 2.3x more concurrent agents and 1.6x greater CPU capacity for robotic workloads.

Figure 7: AMD's revenue contribution forecast



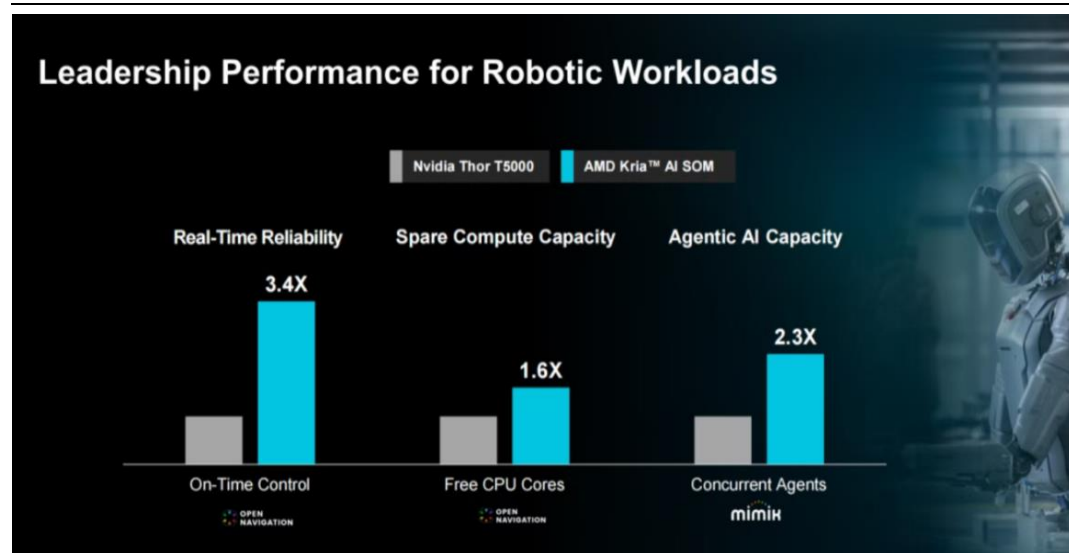
Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 8: AMD's physical AI TAM estimate



Sources: Company data, GF Securities (Hong Kong) Brokerage

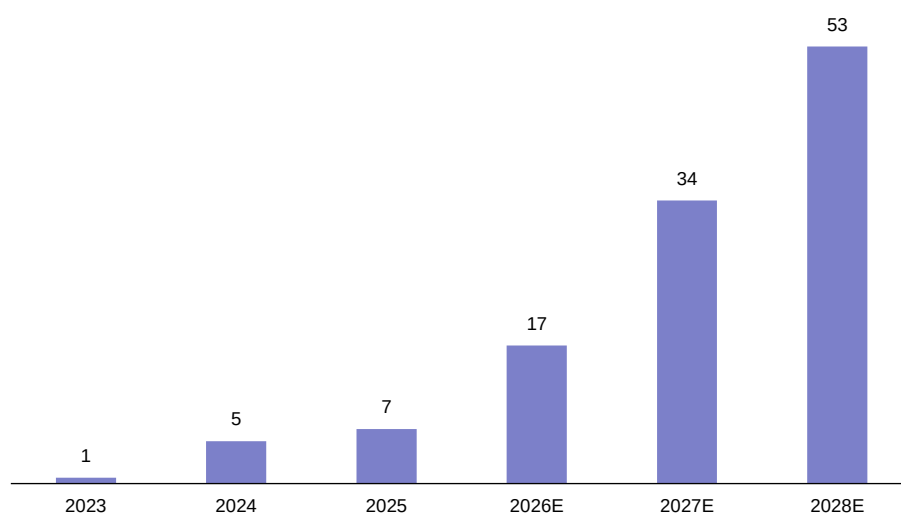
Figure 9: AMD Kria AI vs. Nvidia Jetson Thor



Sources: Company data, GF Securities (Hong Kong) Brokerage

Earnings Forecast

Figure 10: AMD's AI GPU revenue (USD bn)



Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 11: Our forecast of key matrix, vs. AMD's guidance

	2026E	2027E	2028E	Guidance (26-28 or 26-30)
By segment YoY				
Client + Gaming	1%	5%	10%	Higher than 10%
DC	104%	85%	47%	Higher than 60%
DC GPU	153%	105%	54%	Higher than 80%
Embedded	8%	4%	2%	Higher than 10%
Corporate level				
Total sales YoY	51%	57%	38%	Higher than 35%
Gross margin	56%	58%	58%	56%-58%
Operating margin	28%	33%	36%	Higher than 35%
EPS (\$)	7.9	14.2	20.2	Higher than \$20
FCF Margin	22%	24%	27%	Higher than 25%

Sources: Company data, GF Securities (Hong Kong) Brokerage

Figure 12: AMD's P&L forecast

(USD m)	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Sales	34,639	10,253	11,414	12,171	18,487	52,324	18,870	18,934	20,313	24,130	82,247
COGS	16,474	4,568	5,027	5,253	8,085	22,934	8,021	8,013	8,688	10,312	35,034
Gross profit	18,165	5,685	6,387	6,917	10,401	29,390	10,849	10,921	11,625	13,818	47,213
opex	10,397	3,145	3,300	3,500	4,600	14,545	4,600	4,800	5,100	5,400	19,900
Operating profit	7,768	2,540	3,087	3,417	5,801	14,845	6,249	6,121	6,525	8,418	27,313
Non-operating profit	(1,105)	(613)	67	67	67	-412	65	65	65	65	260
Pre-tax profit	6,663	1,927	3,154	3,484	5,868	14,433	6,314	6,186	6,590	8,483	27,573
Income tax	(168)	(338)	410	453	763	1,288	821	804	857	1103	3,584
Non-GAAP Net Income	6,831	2,265	2,744	3,031	5,105	13,145	5,493	5,382	5,734	7,380	23,988
Non-GAAP EPS (\$)	4.17	1.37	1.65	1.83	3.06	7.92	3.29	3.21	3.39	4.33	14.22
Diluted shares	1,637	1,650	1,660	1,660	1,666	1,659	1,672	1,678	1,691	1,705	1,687
Margin analysis											
Gross margin	52.4%	55.4%	56.0%	56.8%	56.3%	56.2%	57.5%	57.7%	57.2%	57.3%	57.4%
Operating margin	22.4%	24.8%	27.0%	28.1%	31.4%	28.4%	33.1%	32.3%	32.1%	34.9%	33.2%
Pre-tax margin	19.2%	18.8%	27.6%	28.6%	31.7%	27.6%	33.5%	32.7%	32.4%	35.2%	33.5%
Effective tax rate	-2.5%	-17.5%	13.0%	13.0%	13.0%	8.9%	13.0%	13.0%	13.0%	13.0%	13.0%
Growth (% YoY)											
Sales	34%	38%	49%	32%	80%	51%	84%	66%	67%	31%	57%
Operating profit	28%	43%	244%	53%	103%	91%	146%	98%	91%	45%	84%
Net income	26%	45%	251%	54%	103%	92%	143%	96%	89%	45%	82%
EPS	26%	43%	245%	53%	101%	90%	139%	94%	86%	41%	80%
Product mix (%)											
Data Center	52%	56%	58%	61%	76%	65%	76%	74%	75%	80%	77%
Client	30%	28%	28%	25%	15%	23%	15%	17%	16%	12%	15%
Gaming	8%	7%	7%	5%	3%	5%	3%	5%	4%	3%	4%
Embedded	9%	9%	8%	8%	5%	7%	5%	5%	5%	4%	5%

Sources: Company data, GF Securities (Hong Kong) Brokerage

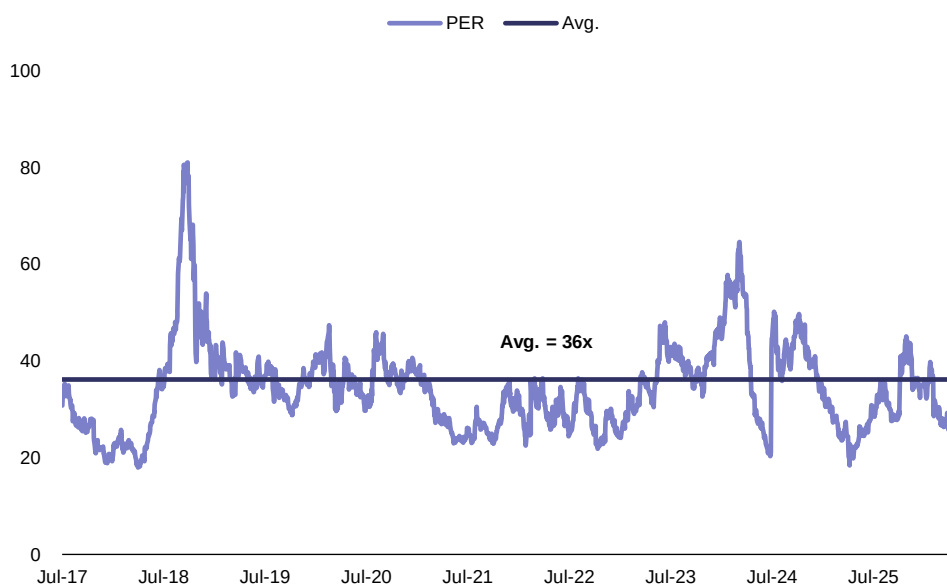
Valuation and Recommendation

Maintain Buy with TP at \$640

We rate AMD (AMD US) Buy and set our TP at \$640, which is based on 45x our 2027E EPS, or 32x 2028E EPS estimate, to reflect its >50% earnings CAGR and enhanced competitive positioning into MI455 and agentic AI product cycle. We adopt P/E methodology as we believe the market will be mainly focused on the company's earnings outlook, and we usually apply P/E method to fabless firms.

AMD's P/E traded at an average of 36x since 2017 and 37x since 4Q22, when the AI cycle started. Our target price multiple is slightly higher than the average level of its AI cycle, despite the fact that the average P/E was distorted by the overly optimistic sentiment of its MI300 cycle in 4Q23-1H24. Further upside or downside to valuation remains the margin trends and how the company executes the MI455 including the rack solution.

Figure 13: AMD's 12-month forward P/E



Sources: Company data, GF Securities (Hong Kong) Brokerage

Risks

1) AI demand slowdown; 2) new product delay; 3) peer competition.

Balance Sheet						USD mn
	2024	2025	2026E	2027E	2028E	
Current assets	19,049	26,947	34,610	47,433	70,172	
Cash and cash equivalents	5,132	10,552	10,783	10,467	19,564	
Inventory	5,734	7,920	11,026	16,843	22,883	
Accounts Receivable	6,192	6,315	9,539	14,994	20,658	
Other current assets	1,991	2,160	3,263	5,129	7,066	
Non-current assets	50,177	49,979	64,791	71,291	77,791	
Long-term equity investment	149	-	11,000	16,000	21,000	
Fixed assets	2,425	2,312	6,124	7,624	9,124	
Other long-term assets	47,603	47,667	47,667	47,667	47,667	
Total assets	69,226	76,926	99,401	118,724	147,963	
Current liabilities	7,281	9,455	16,035	20,978	27,256	
Accounts Payable	2,466	2,929	4,078	6,229	8,463	
Short-term borrowings	-	874	3,420	1,329	304	
Other current liabilities	4,815	5,652	8,538	13,420	18,489	
Non-current liabilities	4,377	4,472	9,042	3,632	976	
Long-term Debt	1,721	2,348	7,980	3,101	710	
Other non-current liabilities	2,656	2,124	1,062	531	266	
Total liabilities	11,658	13,927	25,078	24,610	28,232	
Total Equity	57,568	62,999	74,323	94,113	119,731	
Share Capital	61,379	63,382	62,563	61,367	59,773	
Additional Paid-in Capital	-6,106	-7,079	-8,081	-11,083	-19,085	
Retained earnings	2,295	6,696	19,841	43,830	79,043	
Minority interests	-	-	-	-	-	
Total Liabilities & Equity	69,226	76,926	99,401	118,724	147,963	

Income Statement						USD mn
	2024	2025	2026E	2027E	2028E	
Revenue	25,785	34,639	52,324	82,247	113,313	
Cost of sales	12,026	16,474	22,934	35,034	47,599	
Gross profit	13,759	18,165	29,390	47,213	65,715	
Operating Expense	7,621	10,397	14,545	19,900	25,500	
Operating profit	6,138	7,768	14,845	27,313	40,215	
Interest Income	-	-	0	0	0	
Interest Expense	92	36	-148	-140	-140	
Net other Non-op. Income/(Loss)	-1,436	-1,069	-560	120	120	
Pre-tax profit	4,610	6,663	14,433	27,573	40,475	
Income tax	-810	-168	1,288	3,584	5,262	
Profit for the year	5,420	6,831	13,145	23,988	35,213	
Minority interest	-	-	-	-	-	
Net profit to ord. equity	5,420	6,831	13,145	23,988	35,213	
EPS (\$)	3.3	4.2	7.9	14.2	20.2	

Cash Flow Statement						USD mn
	2024	2025	2026E	2027E	2028E	
Operating cash flow	3,041	7,709	12,868	21,124	32,593	
Profit for the year	5,420	6,831	13,145	23,988	35,213	
Depreciation & amortization	3,064	3,004	3,121	3,240	3,718	
Change in working capital	-685	-3,882	-3,398	-6,105	-6,339	
Others	-4,758	1,756	-	-	-	
Investing cash flow	-1,101	-5,533	-17,933	-9,740	-10,218	
Capex	-3,267	-2,891	-6,933	-4,740	-5,218	
Change in investment	-50	149	-11,000	-5,000	-5,000	
Others	2,216	-2,791	-	-	-	
Free cash flow	1,940	2,176	-5,065	11,384	22,375	
Financing cash flow	-2,062	-431	5,295	-11,699	-13,277	
Change in Capital	-3,744	-1,400	-1,821	-4,198	-9,596	
Net Change in Debt	-1,282	1,501	8,178	-6,970	-3,416	
Others	2,964	-532	-1,062	-531	-266	
Exchange influence	-	-	-	-	-	
Net increase in cash	-122	1,745	231	-316	9,098	

Key Financial Ratios					
	2024	2025	2026E	2027E	2028E
Growth (%)					
Revenue growth	13.7	34.3	51.1	57.2	37.8
Operating profit growth	26.5	26.6	91.1	84.0	47.2
Net profit growth	26.0	26.0	92.4	82.5	46.8
Profitability (%)					
Gross profit margin	53.4	52.4	56.2	57.4	58.0
Operating Profit Margin	23.8	22.4	28.4	33.2	35.5
Net profit margin	21.0	19.7	25.1	29.2	31.1
Key Ratio (%)					
ROA	7.9	9.3	14.9	22.0	26.4
ROE	9.4	10.8	17.7	25.5	29.4
Stability					
Gross debt/equity	3.0	5.1	15.3	4.7	0.8
Interest Coverage	51.1	186.1	-96.5	-195.9	-288.1
Current Ratio	2.6	2.9	2.2	2.3	2.6
Quick Ratio	1.8	2.0	1.5	1.5	1.7
Net debt/equity	Net cash	Net cash	Net cash	Net cash	Net cash

Rating definitions Benchmark: Hang Seng Index (Hong Kong)

Company ratings	Buy	Stock expected to outperform benchmark by more than 10%
	Hold	Expected stock relative performance ranges between -10% and 10%
	Underperform	Stock expected to underperform benchmark by more than 10%

Sector ratings	Positive	Sector expected to outperform benchmark by more than 10%
	Neutral	Expected sector relative performance ranges between -10% and 10%
	Cautious	Sector expected to underperform benchmark by more than 10%

Hong Kong	Company	GF Securities (Hong Kong) Brokerage Limited
	Address	27/F, GF Tower, 81 Lockhart Road, Wan Chai, Hong Kong
	Telephone	(852) 37191111
	Email	evanlee@gfgroup.com.hk

Disclaimer

This report has been prepared by GF Securities (Hong Kong) Brokerage Limited ("GF Securities (Hong Kong) Brokerage"). According to the laws, regulations and regulatory requirements in different countries and regions, this report is distributed by GF Securities (Hong Kong) Brokerage with relevant legal and compliant operation qualifications in these countries and regions.

GF Securities (Hong Kong) Brokerage is licensed by the Securities and Futures Commission of Hong Kong ("SFC") to conduct Type 4 Regulated Activity "Advising on Securities". It is regulated by the SFC, and is responsible for the distribution of this report in Hong Kong. Information about the qualifications of the research analyst(s) who is(are) the author(s) of this report as licensed by the SFC are disclosed in the section where research analyst names are shown.

The research analyst(s) primarily responsible for the content of this report, in whole or in part, certifies that with respect to the company or relevant securities that the analyst(s) covered in this report: 1) all of the views expressed accurately reflect his or her personal views on the company or relevant securities mentioned herein; and 2) no part of his or her remuneration was, is, or will be, directly or indirectly, in connection with his or her specific recommendations or views expressed in this report.

This report is published solely for information purpose and does not constitute an offer to buy or sell any securities or a solicitation of an offer to buy, or recommendation for investment in, any securities.

The securities mentioned in this report may not be allowed to be sold in certain jurisdictions. No action has been taken to permit the distribution of the research report to any person in any jurisdiction that the circulation or distribution of such research report is unlawful. This report is distributed solely to clients or designated institutions authorized by GF Securities (Hong Kong) Brokerage, and is not distributed publicly. It is distributed to certain clients based on the conclusion that they are able to assess investment risks independently, execute investment decisions independently and assume corresponding risks independently.

This report has been issued and based on information obtained from sources generally available to the public and believed by the research analyst(s) to be reliable but which has not been independently verified. No representation or warranty, either express or implied, is made by GF Securities (Hong Kong) Brokerage as to their accuracy and completeness of the information contained in this report. GF Securities (Hong Kong) Brokerage accepts no liability for all loss arising from the use of the materials presented in this report, unless is excluded by applicable laws or regulations. Please be aware of the fact that investments involve risks and the price of securities may be fluctuated and therefore return may be varied, past results do not guarantee future performance. Any recommendation contained in this report does not have regard to the specific investment objectives, financial situation and the particular needs of any individuals. This report is not to be taken in substitution for the exercise of judgment by respective recipients of this report, where necessary, recipients should obtain professional advice before making investment decisions.

GF Securities (Hong Kong) Brokerage may have issued, and may in the future issue, other communications that are inconsistent with, and reach different conclusions from, the information presented in the research report. The points of view, opinions and analytical methods adopted in the research report are solely expressed by the analysts but not that of GF Securities (Hong Kong) Brokerage or its affiliates. The information, opinions and forecasts presented in the research report are the current opinions of the analysts as of the date appearing on this material only which may subject to change at any time without notice. The salesperson, dealer or other professionals of GF Securities (Hong Kong) Brokerage may deliver opposite points of view to their clients and the proprietary trading division with respect to market commentary or dealing strategy either in writing or verbally. The proprietary trading division of GF Securities (Hong Kong) Brokerage may have different investment decision which may be contrary to the opinions expressed in the research report. GF Securities (Hong Kong) Brokerage or its affiliates or respective directors, officers, analysts and employees related to research report business may have rights and interests in securities mentioned in the research report. Recipients should be aware of relevant disclosure of interest (if any) when reading this report.

GF Securities (Hong Kong) Brokerage and its affiliates may be seeking or building business relationships with company(ies) mentioned in this report. Therefore, investors should consider the impact on the independence of this report by GF Securities (Hong Kong) Brokerage and its affiliates due to potential conflicts of interests. Investors should not make any investment decisions based solely on the contents of this report. Investors should make their own investment decisions and bear their own risk. No written or verbal commitment of sharing gains or losses from securities investments in any form shall be effective.

This report may contain and/or describe/present factual historical information on prices of Futures contracts (the "information"). Please note that this information is solely for the purpose of forming part of the argument/grounds/evidence in our research methodology/analysis to support our conclusion on our view of the relevant industry/company mentioned. It does not, by any means (express or implied) to be associated with or constituted as SFC Type 5 Regulated Activity (Advising on futures contracts).

Disclosure of Interests

- 1) The research analyst(s) and his/her associate has not served as an officer of the company(ies) mentioned in this report, and does not have any financial interests in the company(ies) mentioned in this report.
- 2) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any financial interests and has not invested in interests aggregate to an amount equal to or more than (i) 1% of the market capitalization; or (ii) 1% of the issued share capital, or issued units, in the company(ies) mentioned in this report.
- 3) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any market making activities, and has not employed any individual(s) serving as officer(s) of the company(ies) mentioned in this report.
- 4) GF Securities (Hong Kong) Brokerage and/or its affiliates does not have any investment banking relationship in Hong Kong with the company(ies) mentioned in this report in the past 12 months.

Copyright © GF Securities (Hong Kong) Brokerage

Without the prior written consent obtained from GF Securities (Hong Kong) Brokerage, any part of the materials contained herein should not (i) in any forms be copied or reproduced or (ii) be re-disseminated.